

A Lightweight Security Technique For Unmanned Aerial Vehicles Against GPS Spoofing Attack

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Abstract—The use of Unmanned Aerial Vehicles (UAVs) in military and civil applications is increased in recent years. These UAVs are generally equipped with a set of sensors and follow a predefined trajectory. The regions of deployment are usually hostile environments and inaccessible such as disaster zone or military fields. Therefore, to accomplish their mission, the UAVs need to communicate with each other, with the Ground Control Station (GCS) and with the navigation satellite system. Through these infrastructures, the UAVs are prone and vulnerable to various cyber-attacks such as GPS spoofing. This attack can be launched easily by an attacker using a simple transmitter by broadcasting fake and wrong GPS signals. These signals can mislead UAVs that receive them from their initial trajectory. In order to ensure the predefined UAV's mission, detecting GPS spoofing attacks is a real challenge and permits to get a high level of flight security, high reliability, and schedule maintenance in time. In this research, we model this issue using a Bayesian network in order to detect this type of attack, and then, the model analyzes and detects the fake GPS signal data. The preliminary results that our proposal provides are promising in terms of a set of metrics.

Index Terms—Unmanned aerial vehicles, Bayesian network, GPS spoofing attack detection

I. INTRODUCTION

Unmanned aerial vehicles (UAVs) are also named drones. UAVs have been used in several civil and military applications such as disaster management, aerial surveillance, target tracking aerial photography, search and rescue, monitoring livestock, etc. The accomplishment of UAV's mission depends on secure communications between UAVs and with different entities in their network such as the Ground Control Station (GCS). In fact, the UAVs use the Global Positioning System (GPS) coordinates to ensure the mission by following a predefined trajectory. However, the GPS signals received are not always correct and secure due to the presence of attackers in the monitoring areas.

Unlike in military applications, most of the civilian UAVs employ non-secure communication protocols such as the unencrypted and unauthenticated transmission of GPS signals. Therefore, the missions of these UAVs cannot be ensured without high reliability. In fact, UAVs are prone to cyber-attacks that can target the privacy and integrity of information, especially the GPS spoofing attack. In this attack, an attacker transmits fake GPS signals similar to those transmitted by satellites, but at higher power. The attacker can also launch a similar attack called GPS meaconing, in which it modifies

stored GPS signals (which are already received at another location) and then broadcasts them at the target location in order to mislead UAVs.

In this work, we present a novel GPS spoofing attack detection technique for UAVs. The proposal employs the Bayesian paradigm in order to model the system and detect GPS spoofing attacks by analyzing the received GPS signals. The algorithm is performed online and can be easily implemented on real UAVs. Our contributions are summarized as follows:

- We model the detection attack using the Bayesian network. The obtained Bayesian model is based on a set of variables that influence the detection process.
- We identify the new signal data which represent the evidence. These information are propagated using a well-known evidence propagation algorithm called Pearl's message-passing algorithm [1].
- We demonstrate the performance of the proposed algorithm via simulation under various metrics.

The remainder of the paper is structured as follows. Section III presents some preliminaries. Section IV introduces in detail our technique. Then, we describe the results of the conducted experiments in Section V. Finally, in section VI, we give our conclusions.

II. RELATED WORK

At present, few works related to cyber security in UAV civilian networks have been proposed. The authors in [2], proposed a technique of authentication for civil GPS signals. The technique uses probabilistic models that are based on cryptographic authentication with signal timing authentication in order to make the civil GPS receivers more safe against spoofing attacks. O'Hanlon et al. [3] proposed a technique to detect the spoofing attack in civilian applications. The method has been implemented and tested in a real-time system. Psiaki and Humphreys [4] described in details a set of attack methods and defenses such as GNSS spoofing attacks. The authors presented some detection mechanisms and issues associated with these mechanisms. The authors in [5] proposed a method called SPREE that able to detect spoofing attacks. The authors claim that their proposal has been implemented and tested against three scenarios of GPS signal traces. Panice et al. [6] presented a technique in which the Support Vector Machine is used for detecting the GPS spoofing attack in UAVs. The method analyzes the state estimation and exploits distribution

errors that exist between GPS and inertial navigation. The authors in [7], proposed a technique called Crowd-GPS-Sec for detecting and localizing GPS spoofing attacks on UAVs and airliners. The authors claim that their method does not require updating the GPS receivers. The detection of attacks is done by monitoring and analyzing the air traffic. The method computes the differences in position coordinates of a set of UAVs by an independent infrastructure on the ground. Eldosouky et al. [8] introduced a mathematical model to analyze and mitigate the GPS spoofing attacks on UAVs. In this proposal, the model can detect the spoofer's effect on UAVs and let them predict the correct trajectory. In fact, the UAVs calculate their location basing on their neighbors using the Stackelberg game as a model. The authors in [9] presented a method for detecting the GPS spoofing signals on Unmanned Aerial Systems (UAS). The proposed technique employs artificial neural networks in order to classify the received signals which can be sent by an attacker. The obtained results are very promising and present a high probability of detection with a low percentage of false alarms. However, all these works do not consider the online detection of attacks and the high mobility of the UAVs.

III. PRELIMINARIES

A. System model

We denote the finite set of UAV by $\mathcal{UAV} = \{UAV_1, UAV_2, \dots, UAV_n\}$ and the finite set of links by $\mathcal{L} = \{L_{i,j} | UAV_i, UAV_j \in \mathcal{UAV} \wedge i \neq j\}$. We consider non-oriented bi-directional links, i.e., link $L_{i,j}$ is the same as $L_{j,i}$. A link $L_{i,j}$ exists, if UAV_i and UAV_j belong to the same transmission range. Further, we assume that UAVs are equipped with a GPS receiver. The GPS receiver receives the signals from at least four or more satellites and computes its three-dimensional coordinates.

B. Bayesian Network

Most areas of application and use of UAVs are characterized by uncertainty. In fact, the UAVs are generally deployed in remote and inaccessible regions such as military fields or disaster zones. The accuracy of measurements cannot be predicted with complete certainty. The GCS can estimate how likely the accuracy of the UAV's measurements based on the probability computing. This helps in detecting attacks on UAVs. The Bayesian network is a useful model in that it allows predicted decisions to be made using predefined real variables.

A Bayesian network is a Directed Acyclic Graph (DAG) that represents a probability distribution. In such a graph, each random variable X_i is denoted by a node. A directed edge between two nodes indicates a probabilistic influence (dependency) of a child. Consequently, the structure of the network denotes the assumption that each node X_i in the network is conditionally independent of its non-descendants given its parents. To describe a probability distribution satisfying these assumptions, each node X_i in the network is associated with a conditional probability table (CPT_i), which specifies the distribution over X_i given any possible assignment of values to its parents[1].

Let $G = (\psi, \zeta)$ denotes a DAG, where ψ and ζ represent the nodes and the edges respectively. Hence, let $X = (X_i)_{i \in \psi}$ a set of variables associated with ψ . The GCS can predict UAV's failures using the joint probability computed over the Bayesian network. The latter can be expressed as follows:

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^n p(X_i | \text{par}(X_i)) \quad (1)$$

Where X_i represents the variable defined on the network and $\text{par}(X_i)$ represents the parents of the node.

Finally, computing the probability of a conditional variable $p(X_i | X_j)$ can be done using the following Bayes' theorem.

$$p(X_i | X_j) = \frac{p(X_j | X_i) p(X_i)}{p(X_j)} \quad (2)$$

IV. THE PROPOSED GPS ATTACK DETECTION TECHNIQUE

In order to model an efficient defense technique against the GPS spoofing attack, we envision a lightweight Bayesian-based model within UAVs. The designed model can act as an attack detector module integrated within UAVs. This module uses a well-known inference algorithm proposed by Pearl called message-passing algorithm [1]. In fact, when new information has arrived, i.e., when an UAV received new GPS information, the UAV will check its correctness by the propagation of this new information (called *Evidence*) and compute the new conditional probability of the proposed Bayesian network. To do that, the UAV detector module needs to use Pearl's algorithm and extracts the CPT of the last node of the proposed Bayesian network. Next, the UAV can take the final decision about the new GPS information basing on the previous CPT parameter probabilities. Fig. 1 shows the global view of our proposal. Next, we describe in detail how our Bayesian model is built and we define all the required variables.

A. Bayesian-based attack detection model

The overall of this study is to detect the GPS spoofing attack. Therefore, deciding that the received GPS information is sent by an attacker depends on three useful information:: The new GPS information, the current GPS information, and the neighbor GPS information. All these information can be represented by three variables in our Bayesian network. Fig. 2 illustrates our initial Bayesian network. The node AD (Attack Detector) represents the final decision variable, i.e., the UAV can decide whether the received GPS information is safe or not, according to the updates AD's CPT. Note, that the CPT of a variable depends on the CPT of its children. Therefore, the AD's CPT is updated after arriving a new GPS information at the children nodes level using the inference process.

1) *New GPS information (N_GPS_inf) variable:* The N_GPS_inf represents all the information considered as essential for the UAV new location, i.e., information concerning the new position that the UAV should go to. The GPS information can be resumed on the received GPS signal. Four information are considered as significant for a GPS receiver: space vehicle

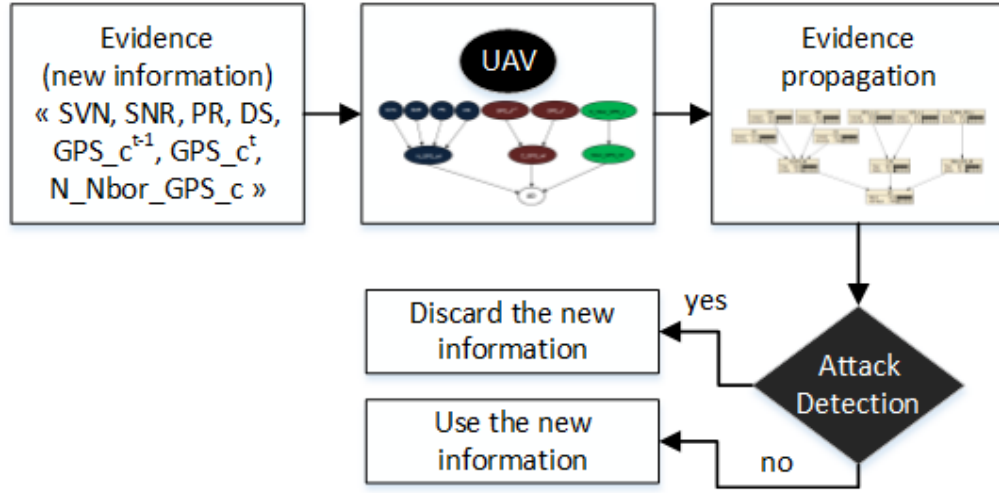


Fig. 1: GPS attack detection process.

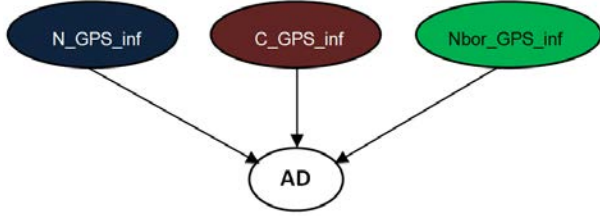


Fig. 2: Our initial Bayesian network .

number, signal-to-noise ratio, pseudo-range, and doppler shift (DS). We represent each of these information by a variable (node) in our Bayesian network.

a) *Space Vehicle Number (SVN)*: is a serial number assigned to a GPS satellite. In fact, the GPS receiver installed in an UAV can extract this number easily from the decoded received GPS signal.

b) *Signal-to-Noise Ratio (SNR)*: is a metric in which the GPS receiver can estimate the signal strength.

c) *Pseudo Range (PR)*: is the pseudo distance between the GPS satellite and the UAV's GPS receiver. To calculate this measurement, the UAV's GPS receiver uses the following equation:

$$\begin{aligned} PR &= c \times \Delta T \\ &= c \times (T_r - T_s) \end{aligned} \quad (3)$$

where:

- T_r : is the time signal reception;
- T_s : is the time signal transmission.

d) *Doppler Shift (DS)*: is the apparent difference between the frequency at signal leaves a source and that at which it reaches a receiver. The DS can be calculated using the following equation:

$$f = f_0(v + vr)/(v + vs) \quad (4)$$

where:

- f : is the observed frequency of the signal;
- f_0 : is the frequency of the emitted signal;
- v is the velocity of the signals in the medium;
- vr is the velocity of the receiver;
- vs is the velocity of the source.

Now, we can apply the equation 1 on the N_GPS_inf variable, we obtain:

$$\begin{aligned} P(N_GPS_inf) &= p(N_GPS_inf|par(N_GPS_inf)) \\ &= p(N_GPS_inf|SVN, SNR, PR, DS) \end{aligned} \quad (5)$$

The sub-graph corresponds to the N_GPS_inf variable is shown by the Fig. 3.

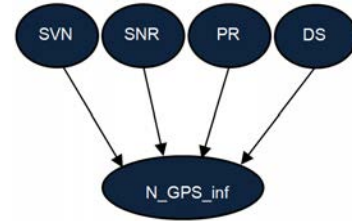


Fig. 3: The N_GPS_inf sub-graph.

2) *Current GPS information (C_GPS_inf) variable*: The second variable that influences the final decision is the C_GPS_inf , i.e., the current GPS information. In fact, this variable refers to the current position of the UAV at time t , before moving to the new position. So, when the UAV moves to a new position, it will calculate the C_GPS_inf probability basing on the latest GPS coordinates and the new one as illustrates in the Fig. 4. While the C_GPS_inf is affected by two variables: GPS_c^{t-1} and GPS_c^t , we can represent the sub-graph of the variable C_GPS_inf such as in the Fig. 5. Now, and in order to compute the joint probability, we apply



Fig. 4: UAV movement process.

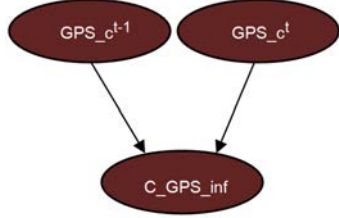


Fig. 5: The C_GPS_inf sub-graph.

the Equation 1 on the C_GPS_inf variable, we obtain:

$$\begin{aligned} P(C_GPS_inf) &= p(C_GPS_inf|par(C_GPS_inf)) \\ &= p(C_GPS_inf|GPS_c^{t-1}, GPS_c^t) \end{aligned} \quad (6)$$

3) Neighbor GPS information ($Nbor_GPS_inf$) variable:

Usually, UAVs move in a fleet manner in order to monitor a region. Therefore, the consecutive positions reached by the UAVs constitute the predefined trajectory. In addition, the UAVs will keep a relative distance between each other. Thus, we have at least for each UAV one neighbor that follows the same trajectory, which leads to the situation where the next position of a neighbor is relatively closed to that UAV. In order, to model this crucial information, we propose the variable $Nbor_GPS_inf$ which depends on the next position of a selected neighbor. To achieve this end, UAVs should exchange two significant information: the Neighbor Current GPS coordinates $Nbor_C_GPS_c$ and the Next Neighbor Current GPS coordinates $N_Nbor_GPS_c$. This exchange should be done before moving to the next location. When an UAV receives these information, it will proceed to the selection of one neighbor using Euclidean distance (Ed) using the following formula:

$$Ed(l_j, l_c) = \sqrt{(x_j - x_c)^2 + (y_j - y_c)^2 + (z_j - z_c)^2} \quad (7)$$

where:

- (x_j, y_j, z_j) describes the Neighbor Current GPS coordinates ($Nbor_C_GPS_c$)
- (x_c, y_c, z_c) describes the Next Neighbor Current GPS coordinates ($N_Nbor_GPS_c$)

The neighbor selection is based on the result of the calculation of the Ed for all neighbors where the neighbor with a lower Ed is considered as a elected neighbor. The second information received, i.e., the $N_Nbor_GPS_c$ of the corresponding neigh-

bor. This information will be used to compute the probability of the variable $Nbor_GPS_inf$ as follows:

$$\begin{aligned} P(Nbor_GPS_inf) &= \\ p(Nbor_GPS_inf|par(Nbor_GPS_inf)) &= \quad (8) \\ p(Nbor_GPS_inf|N_Nbor_GPS_c) \end{aligned}$$

The correspondent sub-graph is shown in the Fig. 6. Finally,

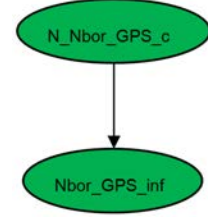


Fig. 6: The $Nbor_GPS_inf$ sub-graph.

our Bayesian network is built based on all the previous sub-graphs, i.e., we merge them on one graph while respecting the causality of the variables. Fig. 7 shows the global Bayesian network. In addition, the joint probability of overall the

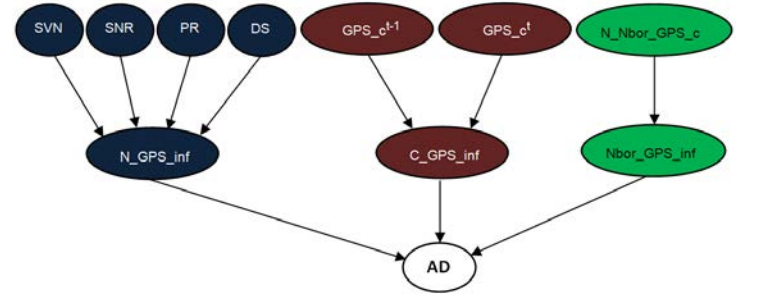


Fig. 7: Our final Bayesian network.

network can be calculated using Equations 1, 5, 7, and 8. We obtain:

$$\begin{aligned} P(AD, N_GPS_inf, C_GPS_inf, Nbor_GPS_inf, SVN, \\ SNR, PR, DS, GPS_c^{t-1}, GPS_c^t, N_Nbor_GPS_c) \\ = P(AD|N_GPS_inf, C_GPS_inf, Nbor_GPS_inf) \\ P(N_GPS_inf|SVN, SNR, PR, DS) \\ P(C_GPS_inf|GPS_c^{t-1}, GPS_c^t) \\ P(Nbor_GPS_inf|N_Nbor_GPS_c) \\ P(SVN)P(SNR)P(PR)P(DS) \\ P(GPS_c^{t-1})P(GPS_c^t)P(N_Nbor_GPS_c) \end{aligned} \quad (9)$$

Now, our model can detect GPS spoofing attack by the propagation of the new information (sent by an attacker) and computing the $P(AD)$ for all its parameters (True and False) by applying the Bayes' theorem given in Equation. 3:

$$\begin{aligned} P(AD) &= P(AD|par(AD)) = \\ P(N_GPS_inf|AD)P(C_GPS_inf|AD) \\ P(Nbor_GPS_inf|AD)P(AD) \end{aligned} \quad (10)$$

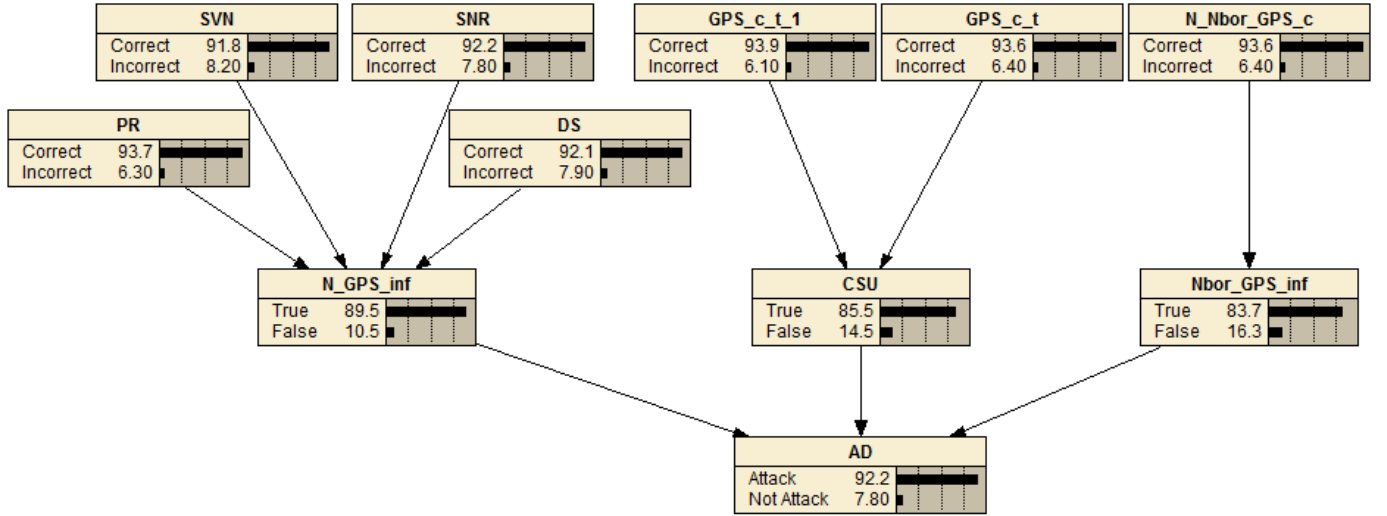


Fig. 8: Our CPT's Bayesian network.

TABLE I: The obtained conditional probabilities using the SatGrid:G22.

ID Variable	Parameter	Prior Value
SVN	Correct	91.8%
	Incorrect	8.2%
SNR	Correct	92.2%
	Incorrect	7.8%
IAS	Correct	96.7%
	Incorrect	3.3%
PR	Correct	93.7%
	Incorrect	6.3%
DS	Correct	92.1%
	Incorrect	7.9%
GPS_c^{t-1}	Correct	93.9%
	Incorrect	6.1%
GPS_c^t	Correct	93.6%
	Incorrect	6.4%
$N_Nbor_GPS_c$	Correct	93.6%
	Incorrect	6.4%

V. PERFORMANCE EVALUATION

In order to evaluate our proposal, we assume that the attacker can realize a GPS spoofing attack. The aim of the attacker is to induce wrong positions as a target UAV's GPS. We simulated this attack by using the GPS signal data [10]. In fact, we used two datasets: the Genuine dataset (SatGrid:G22), and the counterfeit or spoofing dataset (SatGrid:S7). We made some changes and preprocessing on these datasets in order to normalize and keep only the input data of our model. Next, we train and test the model by the calculation of the conditional probabilities of each variable. We used the frequentist method [11] for discrete variables, while for the continuous variables, the Parzen-Rosenblatt kernel density estimator is employed. Table I shows the obtained conditional probabilities using the SatGrid:G22. The Fig. 8 illustrates the corresponding CPT's Bayesian network.

Table I shows the impact of the propagation of the new

information given the variables "SVN, SNR, IAS, PR, DS, GPS_c^{t-1} , GPS_c^t , $N_Nbor_GPS_c$ " on the variables "AD, N_GPS_inf , C_GPS_inf , $Nbor_GPS_inf$ ". In fact, we extracted the new information from the spoofing dataset (SatGrid:S7). These new information are considered as a GPS spoofing attack, i.e., the new GPS data sent to UAVs. Finally, the detection of such attack depends on the obtained probability of each parameter. The AD variable shows two parameters: 1) attack parameter which means that the new information is sent by an attacker if the probability is very high; 2) not_attack parameter represents that the received information are considered as correct ones.

In order to make a depth evaluation, we proceeded to use the confusion matrix in which the following metrics can be easily computed using the corresponding equations:

$$Precision = \frac{TP}{TP + FP} \quad (11)$$

$$Recall = \frac{TP}{TP + FN} \quad (12)$$

$$FPR = \frac{FP}{FP + TN} \quad (13)$$

TABLE II: The impact of the propagation of new information on the probabilities of the variables

ID Variable	Parameter	Obtained Value
AD	Attack	92.2%
	Not Attack	7.8%
N_GPS_inf	True	87.3%
	False	12.7%
C_GPS_inf	True	91.7%
	False	9.3%
$Nbor_GPS_inf$	True	88.7%
	False	11.3%

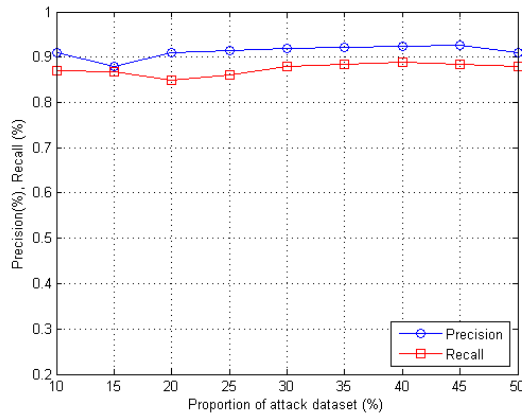


Fig. 9: Precision and Recall curves.

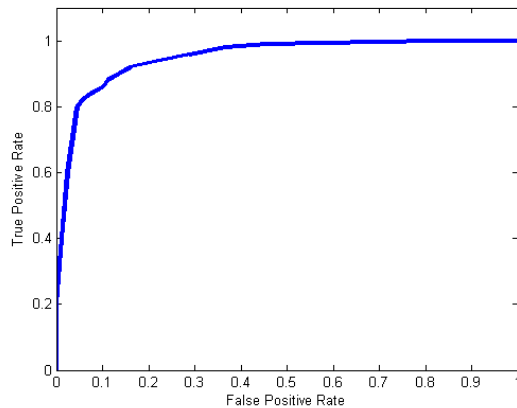


Fig. 10: The ROC curve.

The ROC (Receiver Operating Characteristic) and the AUC (Area Under the ROC) are also calculated. Fig. 9 shows the results of the simulation for precision and recall, respectively. As it can be seen, there is a clear trend that our model presents good results even the portion of the dataset used in simulation increased. These results come from the fact that the model can detect GPS spoofing attacks due to the exploitation of the temporal and spatial GPS information in our Bayesian model.

Fig. 10 demonstrates the ROC curve obtained with regards to the spoofing dataset. This metric is useful for summarizing the attack detection (anomaly detection) with consideration of the FPR metric. In addition, we obtained 0.962 for the AUC metric. The value is considered as an acceptable result.

VI. CONCLUSION

The GPS spoofing attack is considered such a dangerous threat where an UAV can be easily misled. In this paper, we introduced a Bayesian model-based GPS spoofing detection technique for UAVs. This model can be implemented within

UAVs in the pre-deployment phase. In fact, the proposed model analyzes the GPS signal data received by UAVs (SVN, SNR, PR, DS) in order to detect fake signals. The neighbor coordinates are also considered in our model which allows the exploitation of spatial correlation. In addition, the last coordinate of an UAV and the current position are also exploited which represent the temporal correlation. The preliminary simulations demonstrate promising results in which we used a repository of GPS signal spoofing traces called SatGrid:S7 dataset, as well as the Genuine dataset (SatGrid:G22). Future work will include deep experiences where we will introduce several metrics and compare with some works from the literature.

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